GOPINATH R

+91 82486-60104 ▪ Chennai, Tamil Nadu
gopinathr0720@gmail.com ▪ linkedin.com ▪ github.com

CAREER SUMMARY

Senior Software Engineer with 4+ years of experience building scalable backend systems in C#/.NET, with a growing specialization in AI-driven applications — RAG pipelines, vector search, and LLM integrations using AWS, FastAPI, and PostgreSQL. Proven track record modernizing legacy systems, improving application performance, and shipping intelligent features that measurably increase user engagement.

EXPERIENCE

Senior Software EngineerAug 2022 – Present
Triad Software Pvt. Ltd.Chennai, TN

- Built an AI-powered content recommendation engine using **Hugging Face embeddings** and vector search, improving user engagement by **25%**.
- Developed enterprise-grade REST APIs in C#/.NET for licensing, user lifecycle management, and file management features, ensuring secure and scalable backend services.
- Migrated a legacy module from **Cassandra to PostgreSQL**, redesigning the Data Access Layer and WCF services while ensuring seamless data migration and reliability.
- Developed unit tests for **15+ critical database tables** and contributed to performance optimization, reducing production issues and improving deployment quality.
- Built an internal dependency analysis tool that automated method-call hierarchy mapping, reducing regression impact analysis from hours to seconds.
- Collaborated with cross-functional teams in an Agile environment to deliver scalable backend solutions, optimize databases, and participate in code reviews, debugging, and release planning.

Tech Stack: C#, .NET Framework, ASP.NET Core, PostgreSQL, Cassandra, SQL, Python, REST APIs, WCF, Hugging Face, Vector Search, AWS, Git, Agile/Scrum.

Project InternMar 2022 – May 2022
Wipro Technologies. Bangalore, KA

- Developed a Java/J2EE-based web application for online food ordering with end-to-end order management and secure payment integration.
- Designed and implemented backend modules using Java, JSP, Servlets, and Oracle Database following the MVC architecture.
- Built database schemas, wrote SQL queries, and integrated frontend components with backend services.

PROJECTS

AI Mentor
Python | FastAPI | React | Claude SDK | LangChain | MongoDB Atlas | Voyage-3 Embeddings | AWS | Docker
GitHub: github.com/gbnathworkspace/aimentor | Live Demo: mentorman.co.in

- Designed a three-layer memory architecture (L1 profile, L2 skill graph, L3 episodic summaries) to solve context loss across sessions and LLM token limits — persists user goals, per-topic skill level, and prior conversation state without re-feeding full chat history on every turn..
- Built a RAG pipeline using Voyage-3 embeddings and MongoDB Atlas **Vector Search** (cosine similarity, metadata-filtered by user/source/topic) to convert PDFs and cross-topic session summaries into semantically searchable context, retrieved on-demand via agentic tool calls instead of static injection..
- Implemented function/tool calling with Claude via **LangChain**, enabling the model to autonomously pull skill data or search documents/past sessions mid-conversation; bounded the loop to a fixed round cap as a **guardrail** against runaway latency and API cost.
- Developed and deployed scalable backend APIs with FastAPI, containerized with Docker, and hosted on AWS.

SKILLS

Languages: C#, Python, SQL, TypeScript, Java
Frameworks: ASP.NET Core, .NET Framework, FastAPI, React
AI: RAG, Hugging Face, Vector Search, Embeddings, Claude SDK, LangChain
Databases: PostgreSQL, Cassandra, MongoDB
Cloud: AWS, Docker, Git, GitHub Actions
Concepts: Clean Architecture, Microservices, REST APIs, System Design, OOP, Agile, Unit Testing

EDUCATION

B.Tech in Information Technology | 2018 – 2022
Vellalar College of Engineering & Technology. Erode, TN